52. HACKATHON TEAM FORMATION AND RANKING

1. AIM

To collect round scores for hackathon teams, compute total scores, and rank/determine the winning team.

2. PROBLEM STATEMENT

During a college hackathon, multiple teams compete across two rounds. Write a Python program that accepts team names and scores for round 1 and round 2, stores them in tuples, and determines the overall winning team with the highest total score.

3. OBJECTIVES

- 1. Accept number of participating teams.
- 2. Use a list to store team tuples (name, total score).
- 3. Iterate through teams to find the maximum total score.
- 4. Print each team's score and display the winning team.

4. ASSUMPTIONS & REQUIREMENTS

Assumptions:

- Scores are non-negative integers.
- Teams have unique names.

Requirements:

- Python 3.x Interpreter
- Basic List and Tuple data structures

5. ALGORITHM

Step 1: Initialize an empty list teams = [].

Step 2: Read number of teams n.

Step 3: Loop n times: Read team name, r1 score, r2 score. Calculate total = r1 + r2. Append (name, total) to teams.

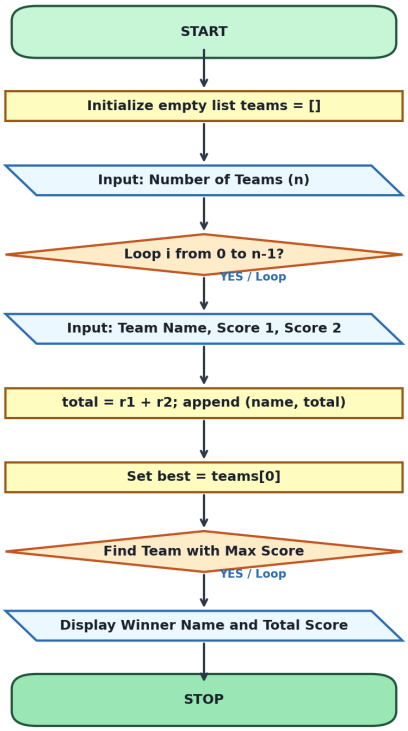
Step 4: Loop through teams list and display each team total.

Step 5: Set best = teams[0]. Iterate teams: if team[1] > best[1], update best = team.

Step 6: Display Winner name and Total score.

Step 7: Stop.

6. PROGRAM FLOWCHART



7. PYTHON PROGRAM (team.py)

```
teams = []
n = int(input("Enter number of teams: "))

for i in range(n):
    name = input("Enter team name: ")
    r1 = int(input("Enter round 1 score: "))
    r2 = int(input("Enter round 2 score: "))
    total = r1 + r2
    teams.append((name, total))

for team in teams:
    print(team[0], "Total:", team[1])

best = teams[0]
for team in teams:
    if team[1] > best[1]:
        best = team

print("Winner:", best[0])
print("Total score:", best[1])
```

8. EXPLANATION OF THE CODE

- **teams = []:** List used to store tuples of (team_name, total_score).
- **teams.append((name, total)):** Appends a 2-element tuple to the list.
- **for team in teams:** Iterates over team tuples to display individual results.
- **best = teams[0]:** Initializes max score tracking with the first team.
- **if team[1] > best[1]:** Compares total score of current team with best score.

9. TRACING TABLE

Team Name	Round 1	Round 2	Total Score	Best Team So Far
whatiswhat	5	4	9	whatiswhat (9)
whyiswhy	4	9	13	whyiswhy (13)

10. TEST CASES

Test Case	Input	Expected Result	Actual Result
TC 1 (Normal)	2 teams: whatiswhat (5,4) whyiswhy (4,9)	Winner: whyiswhy Total score: 13	Winner: whyiswhy Total score: 13
TC 2 (Single Team)	1 team: Alpha (8,8)	Winner: Alpha Total score: 16	Winner: Alpha Total score: 16
TC 3 (Tie Case)	2 teams: TeamA (5,5) TeamB (6,4)	Winner: TeamA Total score: 10	Winner: TeamA Total score: 10

11. EXPECTED OUTPUT

```
Enter number of teams: 2
Enter team name: whatiswhat
Enter round 1 score: 5
Enter round 2 score: 4
Enter team name: whyiswhy
Enter round 1 score: 4
Enter round 2 score: 9
whatiswhat Total: 9
whyiswhy Total: 13
Winner: whyiswhy
Total score: 13
```

12. ACTUAL OUTPUT

```
Enter number of teams: 2
Enter team name: whatiswhat
Enter round 1 score: 5
Enter round 2 score: 4
Enter team name: whyiswhy
Enter round 1 score: 4
Enter round 2 score: 9
whatiswhat Total: 9
whyiswhy Total: 13
Winner: whyiswhy
Total score: 13
```

13. PROGRAM EXECUTION SCREENSHOT

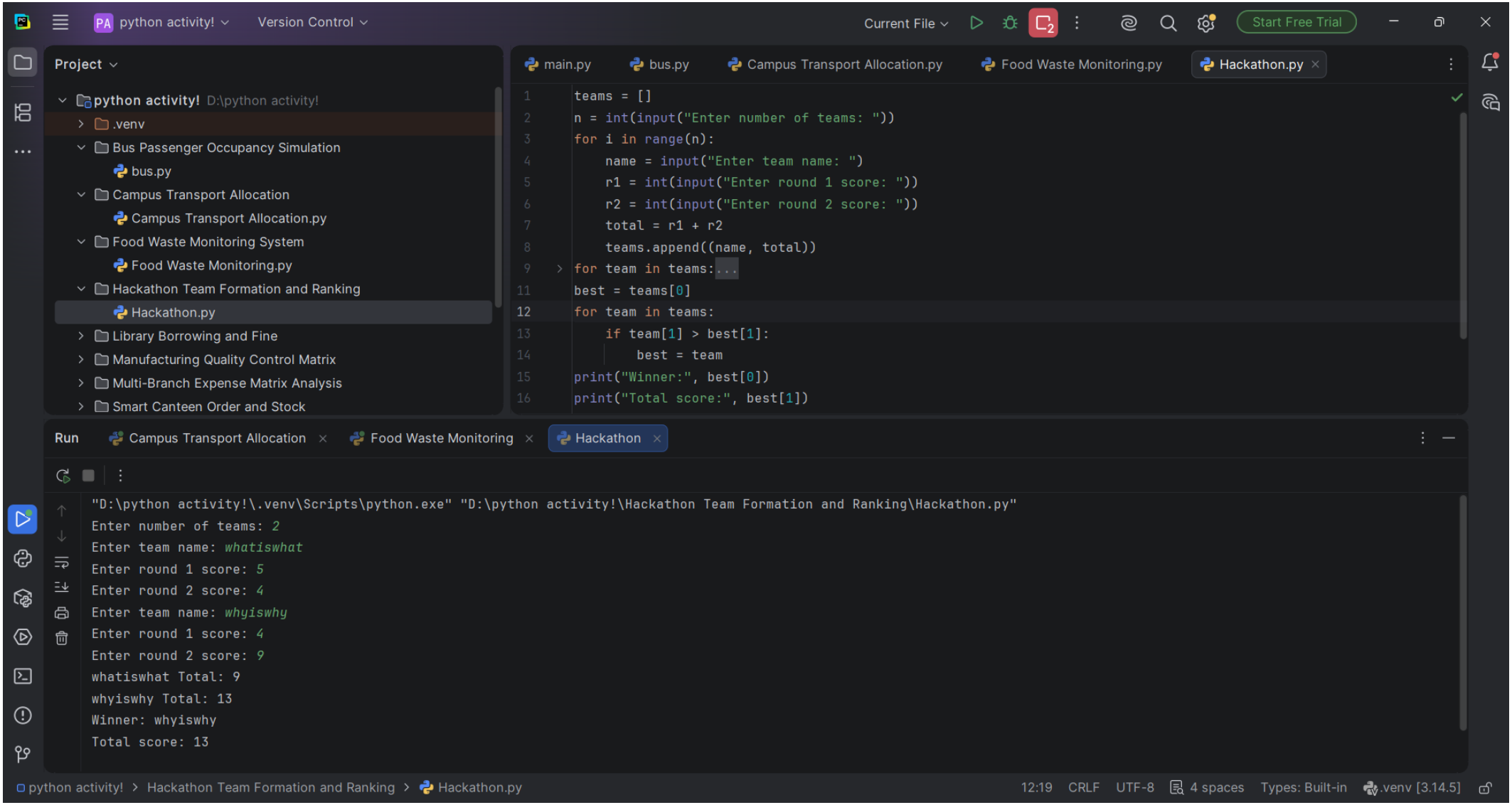
[Screenshot to Be Added]

14. RESULT / CONCLUSION

The Python program team.py was executed successfully. It accurately computes total team scores and identifies the winner.

52 Hackathon Team Ranking

Actual Input and Output – PyCharm Run Screenshot



Screenshot includes the program code and the console inputs/outputs.